

Can LLMs Reliably Count?

Subject–Predicate Distance as a Testbed for Structural Rules in Legislative Drafting Assistants

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Abstract

Legislative drafting assistants such as **SPRINT** use LLM rule checkers to enforce technical requirements (e.g. “§ generally contains at most six paragraphs”). Many rules require precise counting or structural measurement. We propose a narrow benchmark: can LLMs identify the Czech subject (*podmět*) and predicate (*přísudek*) in legal sentences and compute their linear token distance? Using KUK 1.0 (UDPipe silver parses on legal/admin texts) and CLTT 2.0 (manual gold) we compare three prompting regimes (raw text, raw-to-CoNLL-U, pre-parsed CoNLL-U) across model sizes. Results directly guide SPRINT’s hybrid design: counting-heavy rules need deterministic parsers or structured tool-use rather than pure LLM prompts. We address silver-standard reliability and advocate a lightweight human validation layer.

1 Introduction

SPRINT (“IDE + linter + static analysis for law”) turns legislative drafting into a software-engineering workflow: lawyers write *rules as code* (with test sets and versioning), and an LLM evaluator checks compliance at sentence, paragraph, section or document level [1]. The underlying metafora is explicit: “Czech legislation is a large codebase”; a new act is a *diff*, cross-references form a dependency graph, and many drafting rules are *architectural fitness functions* or *linter rules*.

A non-trivial subset of LPV rules requires **exact counting or structural measurement** (e.g. § 39(2): a § generally has at most six paragraphs; future cognitive-complexity metrics based on reference nesting or subject–predicate separation). LLMs are known to fail at precise counting and long-range arithmetic even with chain-of-thought [4, 5]; legal Czech (long sentences, flexible word order, nominal style) amplifies the problem. Pure LLM prompting therefore risks silent errors on rules SPRINT is intended to enforce reliably.

We therefore propose **subject–predicate distance prediction** as a minimal, objectively evaluable proxy task that (a) requires accurate syntactic role labelling *and* exact counting, (b) is directly computable from a UD parse, and (c) maps onto SPRINT needs. We detail the design, discuss silver vs. gold standards on KUK/CLTT, and give concrete recommendations for hybrid SPRINT architecture.

2 Background and Related Work

Legislative rules that count. The Czech *Legislativní pravidla vlády* (LPV) and accompanying methodological guidelines contain dozens of constraints that are, in software terms, `assert` statements over counts, sets or graph properties. SPRINT already implements a growing catalogue of micro-level (sentence/paragraph) linter rules; macro-level and system-level rules (document structure, cross-reference consistency, transposition coverage) are under development. Any rule whose verdict depends on “how many” or “how far” is a candidate for the benchmark proposed here.

LLM limitations on counting and structure. Recent analyses show that transformers struggle with length generalisation, exact arithmetic and symbolic manipulation even when the underlying facts are present in context [4, 5]. Providing structured intermediate representations (e.g., CoNLL-U) can dramatically improve performance on downstream reasoning tasks [6]. Our design explicitly measures this gap for a legal-Czech structural task.

Resources for Czech legal parsing. UDPipe 2.0 [2] supplies the automatic annotations in KUK 1.0, a 6k-document corpus of Czech legal and administrative texts created for comprehensibility research. The Czech Legal Text Treebank 2.0 (CLTT 2.0) [3] offers manually annotated legal sentences and is the natural gold-standard reference for parser quality in the domain.

3 Task: Subject–Predicate Distance in Czech Legal Sentences

3.1 Formal Definition

Given a sentence $S = w_1 \dots w_n$ from a Czech legal text, let p be the token index of the main-clause subject (*podmět*: typically the `nsubj` or `csubj` dependent of the root or main verbal head) and q the token index of the main predicate (*přísudek*: the verbal head that p depends on, or the root in copular constructions). The **distance** is defined as:

$$d = |p - q| - 1$$

(i.e., number of intervening tokens). Distance is measured on the *linear* token sequence, not on the dependency tree depth; this mirrors cognitive load experienced by a human reader.

The task for an LLM (or a hybrid system) is therefore twofold:

1. Correctly identify the pair (p, q) —or at least their lemmas / surface forms—in the presence of relative clauses, coordination and long adjuncts typical of legal Czech.
2. Return the exact integer d .

3.2 Datasets

KUK 1.0 6 275 legal and administrative texts, 19.5 M tokens, each accompanied by UDPipe 2.0 CoNLL-U files and NameTag NE annotations. Silver-standard parses; ideal scale for the benchmark.

CLTT 2.0 1 121 manually syntactically annotated sentences from Czech legal acts. Gold standard for parser evaluation and for quantifying UDPipe error on the legal domain (especially nsubj attachment and root identification).

A stratified sample of 200–500 sentences with exactly one clear main-clause nsubj–verbal-head pair will constitute the evaluation set. Sentences with coordination of subjects or multiple finite verbs will be filtered or annotated with a “primary” pair by human judges.

3.3 Silver vs. Gold: A Central Methodological Question

Because KUK 1.0 relies on automatic UDPipe parses, any LLM that merely “copies” the silver labels will appear artificially strong. We therefore propose a three-tier validation strategy:

1. **Parser quality on legal text.** Run UDPipe on the CLTT 2.0 texts and report LAS, UAS and, crucially, F1 for nsubj identification and agreement of computed distance with the manual gold distance (tolerance ± 1 and exact). This quantifies the silver-standard noise floor.
2. **Small-scale human gold.** Randomly sample 100 sentences from KUK, obtain two independent human annotations of (podmět, přísudek, distance). Compute inter-annotator agreement (Cohen’s κ on roles, exact/ ± 1 match on distance) and disagreement rate with UDPipe. This gives a realistic upper bound on achievable system performance.
3. **Scale with silver + confidence intervals.** For the full benchmark use the silver labels but report results together with the parser-error estimates obtained in (1)–(2). High-stakes rules in SPRINT can later be restricted to the subset of phenomena where silver–gold divergence is negligible.

We expect UDPipe to be “good enough” (LAS > 85–90 on legal Czech, nsubj F1 high) but not perfect; the human layer is cheap (legal experts already annotate for SPRINT rule test-sets) and essential for credibility.

4 Experimental Design

4.1 Prompting Regimes (exactly as motivated by SPRINT needs)

We compare three families of prompts, all in Czech (to match the target domain) and with temperature 0 for reproducibility:

1. **Raw-text end-to-end.** “V následující větě z právního předpisu identifikuj hlavní podmět a hlavní přísudek. Spočítej přesný počet slov mezi nimi. Odpověz přesně ve formátu: Podmět: <slovo> (pozice <číslo>), Příisudek: <slovo> (pozice <číslo>), Vzdálenost: <číslo> slov.”
2. **Raw text $\rightarrow$ CoNLL-U $\rightarrow$ distance.** The model is first asked to produce a CoNLL-U representation (few-shot with two legal examples), then to compute distance from its own parse. This tests whether intermediate structured reasoning helps.
3. **Pre-parsed CoNLL-U input.** The gold/silver CoNLL-U is supplied in the prompt; the model only has to locate the nsubj row, follow its head and count the linear distance. This isolates the “reading structured data + arithmetic” capability.

4.2 Models

- Small: Llama-3-8B, Mistral-7B-Instruct, or a Czech-adapted 7–9B model available via UFAL endpoint.
- Large: Claude-3.5/4 Sonnet (the current SPRINT default), GPT-4o, Llama-3.1-405B.

All prompts are run with identical few-shot examples (2–3 legal sentences with gold explanations) and the same output format instruction.

4.3 Metrics

- **Role identification F1:** exact match on surface form or lemma of subject and predicate (primary key = gold token id).
- **Distance accuracy:** exact match, and relaxed ± 1 , ± 2 token tolerance.
- **Mean Absolute Error (MAE)** on distance.
- **Consistency:** standard deviation across three independent runs (temperature 0 should be near zero; any variance indicates prompt brittleness).

Error analysis will categorise failures: wrong clause attachment, off-by-one counting, mis-tokenisation of multi-word legal terms, hallucinated CoNLL-U columns, etc.

5 Pilot Observations and Implications for SPRINT

Although a full-scale run is future work, a qualitative pilot on ten representative legal sentences (hand-selected for length and embedding depth) already reveals the expected pattern:

- On short sentences (≤ 12 tokens) all models succeed in all three regimes.
- On long sentences with two relative clauses the raw-text regime frequently selects a verb inside a relative clause as the “main” predicate or miscounts intervening tokens by 1–3. Large models are better but still produce occasional off-by-one or attachment errors.
- Supplying CoNLL-U raises accuracy dramatically even for the 8B models; the bottleneck shifts from “understanding syntax” to “reading a table and doing simple arithmetic”—a task at which current LLMs are far more reliable.

Direct consequence for SPRINT architecture. Rules whose verdict depends on a count or a structural metric should *not* be implemented as pure LLM prompts. Instead SPRINT should expose a “structural feature extractor” (UDPipe or a fine-tuned legal parser) whose output (distances, depths, reference graph) is either (a) consumed by deterministic Python checks or (b) injected into the LLM prompt as structured context. The CoNLL-U-input regime above is exactly the prompting style that would be used in the latter hybrid setting.

The same principle extends to other counting-heavy future rules: number of recitals, completeness of transposition tables (already prototyped via RAG+alignment in SPRINT), legisvakance calculations, etc. In each case the reliable primitive is a deterministic or high-precision tool; the LLM supplies the higher-level legal interpretation.

6 Conclusion and Roadmap

We have argued that “can LLMs count?” is not an academic curiosity but a concrete engineering constraint for any LLM-based legislative drafting assistant. The subject–predicate distance task is a minimal, fully automatable probe that exposes the gap between raw-text LLM performance and the reliability required by LPV rules. By coupling KUK-scale silver data with a cheap human validation layer on CLTT and a small KUK sample, we obtain a credible benchmark that can be iterated quickly.

Immediate next steps (F1–F2 of SPRINT roadmap):

1. Implement the three-regime evaluation harness and release prompts + sampled data.
2. Quantify UDPipe–human divergence on nsubj/distance in legal Czech.
3. Add a deterministic “structural linter” module to SPRINT that computes paragraph counts, reference depths and (optionally) subject–predicate statistics; expose the features to rule authors.
4. For any new counting rule, first attempt a deterministic implementation; fall back to LLM+structured-context only when the phenomenon genuinely requires semantic judgment.

The broader message for the legal NLP community is that hybrid tool-use architectures are not optional niceties but a necessary consequence of the fundamental limitations of current LLMs on precise symbolic and counting operations—limitations that become safety-critical when the output is draft legislation.

References

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